Which Side Is Wrong? Neutral Contradiction Flags Do Not Help Retrieval-Augmented Generation Against Misinformation

Anonymous ACL submission

Abstract

Retrieval-augmented generation systems increasingly detect contradictions among retrieved passages and flag them to the generator. We ask what such a flag should say, and what the detector must get right for it to help. In pre-registered experiments on two misinformation benchmarks, RAMDocs and ConflictBank, we vary the detector (heuristic, NLI, LLM, random, oracle) and the format (a neutral flag, a directional flag naming the likely misinformation, or removal of the suspect passage), and add synthetic detectors of controlled precision, recall and direction accuracy. Even a perfectly accurate neutral flag does not help. It lowers accuracy on conflict questions for all three generators we test (gpt-4o-mini, gpt-4.1, Qwen2.5-7B), significantly in all three on ConflictBank, and adding an instruction to drop unreliable passages only brings it back to no-flag accuracy. The same flag helps on the genuine conflicts of WikiContradict. Naming the misinformation is what helps: with the same oracle pairs, a directional flag beats the neutral one in every setting (by 17.7–27.9 points over no flags on ConflictBank), and removal gains most. A wrong direction does as much harm as a neutral flag or more, and false flags make removal delete correct passages. With a real LLM detector, a one-call directional flag beats Astute RAG and TrustRAG on RAMDocs and ties them on ConflictBank, where they do well. We release the harness, the pre-registration and every run.

1 Introduction

Retrieval-augmented generation (RAG) (Lewis et al., 2020) hands a language model passages that may disagree. Some disagreements are genuine: two sources report different findings, or two people share a name. Others are planted or stale misinformation (Zou et al., 2025; Wang et al., 2025b). A natural fix is to detect the contradiction and tell the generator about it. Recent systems pair pairwise contradiction detectors with prompts that flag conflicting passages (Wang et al., 2026), and prompting a model about conflicts can help (Cattan et al., 2025). These systems report detection F1 and end-to-end accuracy, but not how the two connect. What should the flag say, and what must the detector get right for it to help?

We separate the two. We fix the retrieved passages and vary, independently, the *detector* (a heuristic, an NLI cross-encoder, an LLM, a random placebo and an oracle) and the *format* that passes its output to the generator: a neutral "these passages may conflict" flag, a *directional* flag that names the likely misinformation, or removal of the suspect passage. To see what detector quality buys, we also build synthetic detectors with a chosen precision, recall and direction accuracy. After pilots on 200 RAMDocs questions (Wang et al., 2025b), we pre-registered our hypotheses and tested them on 300 held-out questions with three generators and two prompt skeletons. A second registration added ConflictBank (Su et al., 2024), which had no pilot, two published conflict-robust methods, and a neutral flag with an explicit resolution instruction. WikiContradict (Hou et al., 2024), whose conflicts are genuine, is the contrast case. Our claims concern injected misinformation scored by strict exact match; we do not claim they carry over to open-ended answers or to conflicts in which both sides are valid.

Our findings:

- **A correct neutral flag does not help against misinformation.** Flagging exactly the true contradictions lowers gpt-4o-mini's accuracy on RAMDocs conflict questions by 6.7 points $[-10.8, -3.1]$. The estimate is negative in every setting we ran: three generators and two prompt skeletons on RAMDocs, and three generators on ConflictBank, where all three drops are significant. Telling the model to ignore unreliable

passages only cancels the harm. The same flag *raises* accuracy on WikiContradict, where both sides are valid, by 6.5 points (19.0 with gpt-4.1).

- **Acting on direction helps.** Removing the passage the oracle names as misinformation gains on conflict questions in every setting (10.8–30.2 points on RAMDocs, 63.7–78.3 on ConflictBank). A directional flag beats the neutral flag everywhere. It beats no flag on ConflictBank for all three generators, and on RAMDocs for the generators that follow the warning (gpt-4.1; gpt-4o-mini with the default prompt). Naming the wrong side costs 7.0 points as a flag and 21.8 with removal.
- **Removal is the high-variance choice.** False flags make the model lose correct passages. With a real LLM detector, removal's 18.7-point gain on conflict questions is offset by a 20.6-point loss on clean ones, in every generator.
- **With a real detector, direction is competitive with published methods.** A gpt-4o-mini detector's directional flag, in one generator call, beats Astute RAG (Wang et al., 2025a) and TrustRAG (Zhou et al., 2025) on labelled RAMDocs questions by +10.8 and +23.0 points; their consolidation step loses legitimate same-name answers. On ConflictBank it ties them. A Qwen2.5-7B detector gives the same picture. The heuristic detector we started from does not beat random flags of the same volume, and a grid of synthetic detectors predicts real detectors' effects only roughly.

For conflict-aware RAG, we recommend reporting direction accuracy and effects on conflict-free questions alongside detection F1, under the exact format that feeds the generator. We release the harness, pre-registration and every run.[1]

2 Related Work

Knowledge conflicts. Language models can give priority to memorised knowledge over contradicting context (Longpre et al., 2021). They are also readily persuaded by coherent external evidence that contradicts their memory (Xie et al., 2024), and they adopt wrong context over a correct prior to a degree that depends on how strong that prior is (Wu et al., 2024). Xu et al. (2024) distinguish context–memory, inter-context and intra-memory conflicts. We study inter-context conflicts: disagreements among retrieved passages, as in Chen et al. (2022), who find that models tend to rely on a single passage when the evidence conflicts.

Benchmarks with conflicting evidence. RAMDocs (Wang et al., 2025b) mixes genuine ambiguity (several valid answers), misinformation and noise across retrieved documents, and builds on AmbigDocs (Lee et al., 2024). CONFLICTS (Cattan et al., 2025) annotates search results with a taxonomy of conflict types, each with its own desired behaviour. WikiContradict (Hou et al., 2024) pairs Wikipedia passages that genuinely contradict each other. FaithEval (Ming et al., 2025), ConflictBank (Su et al., 2024) and ConfRAG (Yuan et al., 2026) cover related settings. RAMDocs and ConflictBank label which side is misinformation; in WikiContradict neither is.

Telling the model about conflicts. The closest prior work tells the generator that a conflict exists. Cattan et al. (2025) find that prompting a model to reason about the conflict type helps, and that supplying the gold conflict type helps much more. On WikiContradict, models acknowledge both sides more often when told the passages may conflict (Hou et al., 2024). Ragability (Gross et al., 2026) shows the cost of such a hint: it improves conflict identification on contradictory contexts but hurts it on *consistent* ones, where the hint is false. Our neutral-flag result is different: the hint is true, the input is contradictory, and accuracy still falls, because the hint does not say which side to trust. Pan et al. (2023b) test detection and "vigilant" prompting against LLM-generated misinformation, and Pan et al. (2023a) call for misinformation-aware QA. Source signals also matter: models weigh retrieved pages' metadata and appearance (Chiang and Lee, 2024), and passing source-credibility signals to a retrieval-augmented fact-checker helps resolve conflicting evidence (Ge et al., 2025). We separate the *content* of the hint from its *accuracy*.

Telling the model which passage is wrong. Closest to our directional format, Hong et al. (2024) train a discriminator that marks perturbed passages and pass its decisions to the generator in the prompt, which makes the generator more robust to counterfactual noise. CrAM (Deng et al., 2025) rescales attention by per-passage credibility,

[1] Code, the pre-registration and all run outputs are in the anonymised supplementary material and will be released publicly on acceptance.

RA-RAG (Hwang et al., 2025) estimates source reliability, and Weller et al. (2024) defend open-domain QA against disinformation attacks. These works show that *some* reliability signal helps, but do not compare it with a neutral warning on the same pairs or measure its cost on conflict-free questions.

Conflict detection, resolution and poisoning. ConflictRAG (Wang et al., 2026) trains a pairwise detector with an LLM fallback and reports pair-level F1 alongside answer gains; its NLI and chain-of-thought baselines resemble two of our detectors. LLMs are imperfect context validators (Gokul et al., 2025), can detect knowledge conflicts but struggle to pinpoint them (Wang et al., 2024), are distracted by irrelevant context (Shi et al., 2023), and prefer some sources over others (Schuster et al., 2026). TCR (Ye et al., 2026) passes conflict signals through a learned soft prompt, EvidentialRAG (Hossain et al., 2026) routes between answering and abstaining, and credibility-aware generation (Pan et al., 2024) fine-tunes models to weigh per-passage credibility, a trained analogue of our directional flags. MADAM-RAG (Wang et al., 2025b), Astute RAG (Wang et al., 2025a), Self-RAG (Asai et al., 2024) Corrective RAG (Yan et al., 2024) and InstructRAG (Wei et al., 2025) handle unreliable passages without an explicit contradiction detector; we compare against Astute RAG and TrustRAG directly. A few injected passages can steer a RAG system (Zou et al., 2025); RobustRAG (Xiang et al., 2024) isolates passages for certifiable robustness, and TrustRAG (Zhou et al., 2025) filters suspected poisoned ones, the analogue of our removal format. We ask instead how a detector's output should be passed to the generator.

3 Framework

A question q comes with retrieved passages $P = (p_0, \ldots, p_{n-1})$. A *detector* returns flagged pairs $F \subseteq \{(i, j) : i < j\}$. A *directional* detector also returns, for each flagged pair, a *suspect* set $S_{ij} \subseteq \{i, j\}$: the side it believes is misinformation. A *format* turns (P, F, S) into the generator's prompt. We vary detectors and formats independently. Every prompt shares a skeleton: passages, question, and an instruction to end with a `Final answer(s):` line. The *default* skeleton asks for every valid answer when the passages support several (RAMDocs contains same-name ambiguity, so a correct answer may need several entities). The *minimal* skeleton drops that policy, and its answer line allows several answers without inviting them (Appendix A).

3.1 Formats

No flags (plain). Passages and question only.

Instruction. Plain plus a generic instruction to check the passages for contradictions, with no detector. We test two wordings. v1 ends "if a disagreement reflects genuinely different valid answers, report each one"; v2 asks the model to judge which passages are reliable and to leave out answers supported only by unreliable ones.

Flag. Each flagged passage gets the header `[POTENTIAL CONFLICT with passage(s): j]`, followed by a list `Passages i and j: possible contradiction`. This is the format of the original CAAG system, except that every detector's flags are rendered identically (no detector-specific type labels or confidence scores), so detectors differ only in *which* pairs they flag.

Directional flag. As Flag, but suspect passages are headed `[LIKELY MISINFORMATION; conflicts with passage(s): j]`, and the list says which passage likely contains misinformation.

Remove (suspect). The suspect passages of every flagged pair are deleted from the prompt. For a pair without a suspect, the second passage is deleted, which is what the original CAAG REMOVE strategy does when there are no retrieval scores.

3.2 Detectors

CAAG (Contradiction-Aware Attention Gating) is a heuristic detector from an earlier, unpublished RAG system, used here with its code unchanged. A pair is scored only if its MiniLM (Reimers and Gurevych, 2019) cosine similarity reaches $\theta_s = 0.65$. Its score is the sum of a negation and contrast asymmetry term, a named-entity mismatch term (differing dates, numbers, money or percentages) and a temporal-marker term, multiplied by the similarity. A pair is flagged if the score reaches $\theta_c = 0.35$. It runs locally and gives no direction.

NLI is `nli-deberta-v3-base` (He et al., 2023), a cross-encoder scored in both directions; a pair is flagged if the larger contradiction probability exceeds 0.5. It gives no direction.

LLM is a single gpt-4o-mini call (temperature 0) that lists contradicting pairs, telling them apart from ambiguity, complementary and temporal relations, and names the side that is more likely misinformation (or "both" or "unclear"), as JSON.

Random flags each pair independently at CAAG's per-pair flag rate on the confirmatory split (6.22%). It is a placebo that matches CAAG's flag volume.

Oracle flags exactly the gold contradiction pairs and names the gold misinformation side. The *inverted oracle*, an adversarial control, flags the same pairs but names the correct side.

3.3 Synthetic detectors

To separate a detector's accuracy from its identity, we build detectors with a chosen pair-level precision p, recall r and direction accuracy d from the gold labels. Each gold contradiction pair is flagged with probability r. False flags are drawn from non-contradicting pairs at a single per-pair rate, set over the whole split so that expected precision is p. That rate is dataset-wide, so conflict-free questions receive false flags too. A true flag names the gold misinformation side with probability d, and otherwise the other side. A false flag has no misinformation side, so its suspect is picked uniformly. Each pair gets one uniform draw for flagging and one for direction, and every cell reuses them. Cells are therefore nested: raising r or d, or lowering p, only adds flags or corrections. This keeps comparisons across cells paired. The grid crosses $p \in \{0.25, 0.5, 0.75, 1\}$, $r \in \{0.25, 0.5, 1\}$ and $d \in \{0.5, 0.75, 1\}$ (36 cells), each rendered as a directional flag and as removal.

3.4 Placing real detectors on the grid

Can the grid predict a real detector's effect from a few summary numbers? We compare two ways of summarising a detector. *Pair-level*: its pair precision, recall and direction accuracy. *Document-level*: what it does to each subset's documents in a given format. For conflict questions, that is the share whose misinformation documents are all marked and the share that have a correct document marked; for clean questions, the share that have a correct document marked. Here "marked" means removed (Remove) or headed as misinformation (Directional flag). The prediction for a subset is the inverse-distance-weighted mean effect of the four nearest grid cells in that space.

4 Experimental Setup

Data. **RAMDocs** (Wang et al., 2025b) has 500 questions, each with 1–12 documents (5.5 on average) typed *correct* (supports a gold answer, possibly for a different same-name entity), *misinfo* (supports a wrong answer) or *noise*. We label a pair a *contradiction* if one side is misinformation and the two answers differ, *ambiguity* if both are correct with different answers, *consistent* if the answers match, and *unrelated* if either is noise. A question is a *conflict* question if it has a contradiction pair, and *clean* otherwise. The released types are imperfect: 163 documents typed *correct* contain a wrong answer, some as injected misinformation, others incidentally. We leave pairs that touch them out of pair-level metrics, and questions whose only sign of conflict is such a document (55 of 500; 35 of the confirmatory split) count only in "All". Treating them as misinformation is a robustness check.

ConflictBank (Su et al., 2024) builds misinformation differently: for a Wikidata fact, an LLM writes one evidence passage for the true object and one for a substituted object. We sample 400 questions round-robin over relations: 300 conflict questions (correct passage, misinformation passage, two distractors from other subjects) and 100 clean ones (correct passage, three distractors). Every question has one gold answer. It had no pilot, so all its results are confirmatory.

WikiContradict (Hou et al., 2024) has 253 questions, each with two Wikipedia passages that genuinely contradict each other; the reference answer is *both* sides. We also run 237 automatically scorable **CONFLICTS** (Cattan et al., 2025) questions (Appendix D).

Pre-registration and splits. We developed the method on a random 200-question pilot split of RAMDocs. Before running anything on the remaining 300 questions (139 conflict, 126 clean, 35 excluded), we committed a pre-registration that fixes the conditions, metrics, hypotheses and corrections. Every RAMDocs result in the main text is on this confirmatory split; pilot results appear only in the appendix. Comparisons that were not registered, including every real-detector row of Table 1, are exploratory. After review we registered an addendum (H10, H11 and four re-analyses) and a second registration (H12–H14), each before run-

ning anything it covers; Appendix D lists every deviation.

Generators and baselines. The main generator is `gpt-4o-mini` (OpenAI et al., 2024) (temperature 0.1, at most 256 output tokens, seeds 0–2). We replicate the key conditions with `gpt-4.1` (3 seeds) and with Qwen2.5-7B-Instruct (Qwen Team et al., 2024) (4-bit, local, seed 0), keeping the LLM detector fixed at gpt-4o-mini. We rerun the oracle and LLM-detector conditions with the minimal skeleton. MADAM-RAG (Wang et al., 2025b) is ported unchanged (three debate rounds). Astute RAG (Wang et al., 2025a) and TrustRAG (Zhou et al., 2025) use their published prompts and defaults; since both return one answer, we also run *multi* variants that ask for every valid answer (Appendix D). We also run the LLM detector on Qwen2.5-7B with gpt-4o-mini generating.

Metrics. On RAMDocs an answer is correct if it includes every gold answer and no wrong answer (Wang et al., 2025b). Our primary metric reads "includes" literally: after normalisation, each gold or wrong answer must appear as a whole-word span of the extracted final answer. A bidirectional variant, which also counts a predicted answer contained in a gold one, is the robustness metric. We chose the one-way metric before the confirmatory run because it is closer to MADAM-RAG's published score (Appendix D), and we compare conditions only within our own harness. On WikiContradict an answer is correct if it contains both sides (bidirectional matching); on ConflictBank we use the RAMDocs metrics.

Statistics. We average each question's score over seeds, then report paired differences from the plain format with cluster-bootstrap 95% CIs (10,000 resamples) and two-sided sign-flip permutation tests (Efron and Tibshirani, 1993; Dror et al., 2018). Holm correction (Holm, 1979) is applied within each pre-registered family. A hypothesis is *robustly supported* if it also holds under the robustness metric. One significance scheme governs the paper: a registered test takes its family's Holm-adjusted p, and every other comparison we report (254 of them) forms one exploratory Holm family. Only results significant under this scheme are called significant or set in bold. A null difference is not read as equivalence without a TOST (Lakens, 2017). With 139 conflict questions, the minimum detectable effect at 80% power is 4–12 points (median 5.8; Card et al., 2020).

Condition	All	Lab.	Conflict	Clean	Tokens
No flags (plain)	29.9	31.3	22.1	41.5	911
Prompt only					
Instruction (v1)	+1.6	+0.8	−1.2	+2.9	948
Instruction (v2)	+3.1	+2.3	+5.3	−1.1	951
Oracle detector					
Oracle, flag	−3.1	−3.5	**−6.7**	0.0	980
Oracle, directional flag	+6.2	+7.0	**+13.4**	0.0	998
Oracle, remove suspect	**+12.8**	**+14.5**	**+27.6**	0.0	825
Real detectors (exploratory)					
Random, flag	+1.0	+1.0	+0.7	+1.3	949
CAAG, flag	−0.4	−0.8	−0.7	−0.8	942
CAAG, remove	−3.3	−3.8	0.0	−7.9	809
NLI, flag	+0.2	+0.3	−0.5	+1.1	1021
NLI, remove	−5.7	−5.0	−2.2	−8.2	592
LLM, flag	−1.8	−1.3	−4.1	+1.9	970
LLM, directional flag	+5.0	+4.9	+12.5	−3.4	983
LLM, remove suspect	+0.3	0.0	**+18.7**	**−20.6**	718
Reference					
MADAM-RAG	+5.7	+5.7	+3.6	+7.9	8196

Table 1: RAMDocs hold-out, gpt-4o-mini, 3 seeds, one-way strict accuracy (primary). *All*: 300 questions, including 35 whose conflict status the label rule leaves open; *Lab.*: the 265 labelled ones (139 conflict, 126 clean). First row: accuracy (%) with no flags. Other rows: paired difference from it in points; **bold** = significant after Holm correction across the table (sign-flip test, $p < 0.05$). Real-detector rows were not pre-registered as tests. Tokens: mean prompt tokens per call (MADAM-RAG: per question, all calls).

Cost. Token counts come from the API's usage reports. All experiments, pilots included, cost $51.75 in API fees (prices checked on 2026-10-27; Appendix D); Qwen ran locally.

5 Results

Table 1 gives every condition on the RAMDocs confirmatory split. Table 5 (Appendix B) gives every pre-registered test, and Figure 2 (Appendix E) plots each condition's change on conflict questions against its change on clean ones.

5.1 A correct neutral flag does not help against misinformation

Flagging exactly the gold contradictions, in the neutral format, *lowers* gpt-4o-mini's accuracy on conflict questions by 6.7 points [−10.8, −3.1], from 22.1% (H2). With gpt-4.1 the drop is 8.6 points (H9). Both hold under either metric. The errors are not refusals: the share of conflict answers that include the misinformation rises from 47.7% to 61.9%, while gold recall barely moves (72.6% to 69.8%). The model lists the misinformation *alongside* the correct answer.

		Plain	Oracle, conflict q.			LLM, directional flag		LLM, remove suspect	
Generator	Skeleton	conflict	flag	dir. flag	remove	conflict	clean	conflict	clean
gpt-4o-mini	default	22.1	**−6.7**	**+13.4**	**+27.6**	+12.5	−3.4	**+18.7**	**−20.6**
gpt-4o-mini	minimal	20.4	**−5.3**	+5.0	**+18.9**	+3.6	**−10.3**	+11.0	**−19.6**
gpt-4.1	default	21.8	**−8.6**	**+18.2**	**+30.2**	+8.4	0.0	**+20.6**	**−24.3**
gpt-4.1	minimal	33.6	**−11.0**	**+12.7**	**+14.9**	–	–	–	–
Qwen2.5-7B	default	19.4	−5.8	+3.6	**+18.0**	+2.9	−7.1	+7.9	**−18.3**
Qwen2.5-7B	minimal	17.3	−5.8	+2.9	**+10.8**	–	–	–	–

Table 2: The main effects across three generators and two prompt skeletons (RAMDocs hold-out, 3 seeds, one-way strict accuracy). *Plain*: accuracy (%) on conflict questions with no flags. Other columns: paired difference from plain in points. *Default* skeleton tells the model to give every valid answer; *minimal* drops that policy (Appendix A). **Bold**: significant after Holm across the table. The LLM detector is gpt-4o-mini for every generator. Qwen2.5-7B-Instruct runs locally (4-bit).

Not an artefact of the prompt. The default skeleton asks for every valid answer, which could invite this behaviour. Under the minimal skeleton the oracle neutral flag still lowers conflict-question accuracy in all three generators (Table 2): by 5.3 points with gpt-4o-mini and 11.0 with gpt-4.1 (both significant; H10), and by 5.8 with Qwen2.5-7B (not significant; 5.8 under the default skeleton, H11). The mechanism changes. Without the answer policy, gpt-4o-mini no longer adds the misinformation (33.3% to 32.4%) but drops the correct answer (gold recall 57.1% to 47.1%). A neutral flag says that something is wrong without saying what, and the model either keeps both sides or loses the right one.

The same flag helps on genuine conflicts. On WikiContradict, where the reference answer is both sides, the oracle flag *raises* the share of answers that give both sides by 6.5 points [+3.0, +10.0] with gpt-4o-mini and by 19.0 [+14.5, +23.8] with gpt-4.1 (H8; Table 12). Whether a neutral flag helps depends on whether keeping both sides is right.

5.2 Acting on direction helps; a directional flag helps when the model follows it

Given the same oracle pairs, naming the misinformation side beats the neutral flag on conflict questions in all six generator–skeleton settings (H3, H9–H11). Against no flags, *removing* the suspect passage gains in every setting, from 10.8 to 30.2 points, all significant. The directional *flag* is less consistent. It gains 13.4 points with gpt-4o-mini [+6.7, +20.4] and 18.2 with gpt-4.1 under the default skeleton, and 12.7 with gpt-4.1 under the minimal one, but the gain is not significant for gpt-4o-mini under the minimal skeleton (+5.0) or for Qwen2.5-7B (+3.6, +2.9). Qwen largely ignores the header: the share of its answers that include the misinformation falls only from 48.2% to 42.4%, against 47.7% to 14.9% for gpt-4o-mini. The failures concentrate on questions with several gold answers (Appendix C.1; exploratory).

A wrong direction (exploratory). An inverted oracle that names the *correct* side lowers gpt-4o-mini's conflict-question accuracy by 7.0 points as a directional flag [−11.5, −2.9], about as much as a neutral flag, and by 21.8 [−28.8, −15.3] with removal. Direction is a strong lever both ways, which makes it an attack surface (Ethics).

5.3 Replication on ConflictBank

ConflictBank had no pilot, and all 12 of its registered tests (H12) are supported (Table 3). On conflict questions the oracle's neutral flag lowers accuracy for all three generators (gpt-4o-mini −8.8, gpt-4.1 −6.3, Qwen2.5-7B −8.3 points). The directional flag raises it (+27.9, +23.0, +17.7), now significantly for Qwen too, and removal raises it most (+66.4 to +78.3). The mechanism is the same as on RAMDocs. With gpt-4o-mini, the share of conflict answers that include the misinformation is 72.8% with no flags, 83.2% with the neutral flag, 41.2% with the directional flag, and 80.9% when the inverted oracle names the correct passage (−8.6 points). The real LLM detector behaves like a noisier oracle: its neutral flag costs 8.4 points, and its directional flag gains +20.0 (gpt-4o-mini) and +17.0 (gpt-4.1) at most one point on clean questions, though only +6.7 (not significant) with Qwen. Its direction accuracy is 68.8%, against 88.3% on RAMDocs. The NLI cross-encoder flags no pair (Appendix C).

		Plain	Oracle, conflict q.			LLM detector, flag / dir. flag			LLM, remove	
Generator	Seeds	conflict	flag	dir. flag	remove	flag	dir. conf.	dir. clean	conflict	clean
gpt-4o-mini	3	21.6	**−8.8**	**+27.9**	**+66.4**	**−8.4**	**+20.0**	−1.0	**+38.8**	−6.7
gpt-4.1	1	13.0	**−6.3**	**+23.0**	**+78.3**	**−6.7**	**+17.0**	0.0	**+48.7**	−6.0
Qwen2.5-7B	1	23.7	**−8.3**	**+17.7**	**+63.7**	**−8.3**	+6.7	−1.0	**+36.0**	−8.0

Table 3: ConflictBank (300 conflict, 100 clean questions; one correct answer each; LLM-written misinformation built from Wikidata facts), default skeleton. *Plain*: accuracy (%) on conflict questions with no flags; other columns: paired difference in points. The LLM detector is gpt-4o-mini for every generator. **Bold**: significant under the paper's single scheme.

5.4 What a detector must get right

Synthetic detectors (Appendix E, Figure 3) show two thresholds. Only perfect direction gives a significant gain over all questions (directional flag +6, removal +13 points at perfect precision). Precision matters mainly for removal: false flags make the model delete correct passages, which costs clean questions 15 points at precision 0.5, while false directional flags cost at most two (H4a: removal is significantly harmful in 6 of 9 distinct false-flag cells, the directional flag in 0). The grid predicts real detectors' effects only roughly (H6).

5.5 Real detectors (exploratory)

The pre-registered F1 ordering (H1; Table 6) holds only in part: NLI beats CAAG (23.8 vs. 11.6), but the LLM detector (26.4) is not significantly better than NLI. Its direction accuracy is 88.3% [83.3, 92.9]. NLI and the LLM detector flag most clean questions (61.9% and 76.2%). In the neutral format no real detector helps; CAAG does not beat random flags of the same volume (H5; Appendix C).

Direction helps with a real detector too. Holding the LLM detector's pairs fixed, the directional flag beats the neutral flag on conflict questions by 16.5 points [+9.6, +23.7] ($p < 0.001$). Against no flags, it gains 12.5 points on conflict questions (raw $p = 0.002$, not significant after Holm) and costs 3.4 on clean ones; over the labelled questions its net effect is +4.9 [+0.8, +9.3]. Removal gains 18.7 on conflict questions and loses 20.6 on clean ones, netting 0.0 [−6.2, +6.0]. With gpt-4.1, the directional flag's gain over labelled questions is +4.4 [+0.6, +8.3]. Real false flags cost clean questions more than synthetic ones, probably because they concentrate on correct–correct pairs. MADAM-RAG gains 5.7 points overall at roughly nine times the prompt tokens.

5.6 Resolution instructions and published baselines

An instruction does not rescue the neutral flag (H13). Adding the v2 instruction ("ignore unreliable passages") to the oracle's neutral flag leaves conflict-question accuracy where it is without flags (RAMDocs gpt-4o-mini −2.4, gpt-4.1 +5.8, ConflictBank +1.6 points; none significant). With the same instruction, the directional flag beats it in all three settings (Table 4), by +15.6, +25.2 and +62.2 points. The instruction removes the neutral flag's harm but adds nothing, because the model still does not know which side to drop. Given the side, it helps a great deal: the directional flag with the instruction gains +63.8 points on ConflictBank, close to removal, and +30.9 with gpt-4.1 on RAMDocs. Instructions alone help in some settings (gpt-4.1 with v2 on RAMDocs conflict questions, +12.2; v1 on WikiContradict, 8.7), but each wording builds in one policy: v2 lowers both-sides answers on WikiContradict by 6.7 points, and neither changes gpt-4o-mini's RAMDocs accuracy significantly (H7).

Against published methods (H14). Astute RAG and TrustRAG resolve conflicts inside the generator, with two and three calls. On RAMDocs both lose accuracy on clean questions (−11.1 and −24.6 points with gpt-4o-mini, multi variants). They lose valid answers: on clean questions with several gold answers, gold recall falls from 69.0% to 47.9% (Astute RAG) and 29.4% (TrustRAG), against 86.4% to 80.3% and 59.1% with one gold answer (exploratory). Consolidating to the best-supported answer is the wrong policy when passages describe different same-name entities. On labelled RAMDocs questions, a single call with the LLM detector's directional flag beats Astute RAG by +10.8 points and TrustRAG by +23.0 with gpt-4o-mini, and with gpt-4.1 beats TrustRAG by +19.6 and Astute RAG by +6.4

Method	RAMDocs, gpt-4o-mini			RAMDocs, gpt-4.1	ConflictBank, gpt-4o-mini			
	Lab.	Conflict	Clean	Lab.	All	Conflict	Clean	Calls
No flags (plain), accuracy	31.3	22.1	41.5	32.8	38.4	21.6	89.0	1
Instruction (v2) only	+2.3	+5.3	−1.1	–	+4.0	+5.9	−1.7	1
Oracle, flag + instr.	−1.8	−2.4	−1.1	+1.5	+0.7	+1.6	−1.7	1
Oracle, dir. flag + instr.	+6.4	+13.2	−1.1	**+14.7**	**+47.4**	**+63.8**	−1.7	1
MADAM-RAG	+5.7	+3.6	+7.9	–	–	–	–	9×tok.
Astute RAG	**−15.6**	−8.4	**−23.5**	–	**+18.2**	**+27.4**	−9.7	2
Astute RAG (multi)	−5.9	−1.2	−11.1	+2.3	–	–	–	2
TrustRAG	**−17.5**	−11.3	**−24.3**	–	**+13.9**	**+19.9**	−4.0	3
TrustRAG (multi)	**−18.1**	−12.2	**−24.6**	−10.9	–	–	–	3
LLM detector, flag + instr.	−2.0	−2.2	−1.9	–	–	–	–	1
LLM detector, dir. flag + instr.	−3.5	+8.4	**−16.7**	–	–	–	–	1
LLM detector, dir. flag	+4.9	+12.5	−3.4	**+8.7**	**+14.8**	**+20.0**	−1.0	1
LLM detector, remove suspect	0.0	**+18.7**	**−20.6**	–	**+27.4**	**+38.8**	−6.7	1

Table 4: Resolution instructions and published baselines. First row: accuracy (%, one-way strict) with no flags; other rows: paired difference from it in points. *Lab.*: the labelled RAMDocs questions; ConflictBank questions are all labelled. RAMDocs gpt-4o-mini and ConflictBank: 3 seeds; RAMDocs gpt-4.1: seed 0. *instr.*: the v2 reliability instruction. *(multi)*: the baseline's final step asks for every valid answer, which RAMDocs needs (every ConflictBank question has one answer). Calls: generator calls per question (LLM-detector rows add one detector call). **Bold**: significant under the paper's single scheme (Section 4).

(not significant). On ConflictBank, which has one answer per question, the baselines gain (Astute RAG +27.4, TrustRAG +19.9 points on conflict questions) and the directional flag ties both (−3.4, +0.8; not significant). Of the methods we tried with a real detector, only the directional flag does no significant harm on either benchmark or question type while gaining on ConflictBank, and it costs one generator call. A Qwen2.5-7B detector gives the same picture (Appendix C), and on CONFLICTS no condition moves accuracy significantly (Table 10).

6 Discussion

Why neutral flags do not help. A neutral flag says that the passages disagree but not which one to trust, so the model falls back on its answer policy: with the default prompt it keeps both sides, without it it more often drops the correct one. Either way a strict answer to a misinformation question loses, while on WikiContradict, where both sides are right, the same flag helps. This sharpens the finding of Cattan et al. (2025) that knowing the conflict *type* helps: for misinformation, the useful part of the type is the direction. It also extends Ragability's observation that false hints hurt (Gross et al., 2026) to true hints that are not specific enough.

When direction helps. Removal helps with every generator because it needs nothing from the model. A directional flag helps only if the model acts on it, which every generator does on ConflictBank and the smaller models do less on RAMDocs questions with several valid answers. An instruction does not replace direction (H13), but it amplifies it. Practitioners should check that a directional flag changes their generator's behaviour before relying on it.

Why the baselines split by benchmark. Astute RAG and TrustRAG keep the best-supported answer. That is right when one answer is correct (ConflictBank) and wrong when passages describe different same-name entities (RAMDocs). A directional flag leaves the answer policy alone and marks only the suspect passage.

Implications for detectors and interfaces. Detection F1 alone does not say what a detector will do downstream: with the same LLM detector, direction is worth 16.5 points over a neutral flag with gpt-4o-mini but less under the minimal skeleton (+6.0) and with Qwen2.5-7B (+5.0). We recommend reporting (i) direction accuracy, (ii) how often correct documents are marked or removed on conflict-free questions, and (iii) effects on conflict-free questions, under the exact format used downstream. Neither interface is safe with today's detectors: the LLM detector's directional flag costs clean questions 3.4 (default skeleton) to 10.3 points (minimal), and removal loses 20.6.

Removal should be gated on a measured clean-question loss rate.

Limitations

Models. We test three generators: two closed OpenAI models and one 7B open-weights model, run at 4-bit precision with one seed. Larger open models and reasoning models may respond differently, especially to the directional header. The LLM detector is a single prompt, run with two models. We compare against two published methods, both reimplemented from their papers or code (Astute RAG has no official code), but not against a trained detector such as ConflictRAG's (Wang et al., 2026).

Data and power. The confirmatory evidence comes from two benchmarks: RAMDocs (300 questions, 139 of them conflict questions) and ConflictBank (400 sampled questions, 300 with conflicts). In both, the misinformation is synthetic, made by editing or generating text around a substituted answer; ConflictBank's passages are LLM-written on both sides. RAMDocs' document labels are imperfect. We exclude 35 hold-out questions whose status depends on documents we could not label without human annotation, and we check that the conclusions survive treating those documents as misinformation. Our comparisons detect effects of about 5.8 points with 80% power, so smaller effects, including small harms on clean questions, may be missed. WikiContradict has genuine conflicts but no misinformation side. CONFLICTS, the most realistic benchmark we use, shows no significant effect for any condition. It was run with one seed, scored only on items with a reference answer, and without the planned LLM judge, so it can neither confirm nor refute our findings. We do not know how they transfer to naturally occurring misinformation, where the false side may be the better written or the more numerous one.

Metric. RAMDocs ships no scorer. Our primary metric reconstructs the published definition and scores our MADAM-RAG port close to the published number on the pilot split, but the published subset is unknown, so this is not a reproduction. The metric is string matching: it misses paraphrases, and partial answers interact with "no wrong answer". We report a more lenient variant throughout and mark any conclusion that depends on the choice.

Direction and the grid. Oracle direction is close to giving the model an answer key, so the oracle rows are upper bounds. In the synthetic grid only perfect direction gives a significant overall gain. The real LLM detector names the right side 88.3% of the time on RAMDocs and 68.8% on ConflictBank. Its directional flag gains significantly over no flags on ConflictBank, but on RAMDocs its net gain over labelled questions is significant only with gpt-4.1. Synthetic false flags are spread uniformly and their suspect side is random, whereas real detectors concentrate their errors. The grid predicts real detectors only roughly, and three detectors cannot show whether document- or pair-level summaries predict better.

Scope. We study flags passed to the generator as text. Learned interfaces such as soft prompts or credibility fine-tuning (Ye et al., 2026; Pan et al., 2024), and retrieval-side filtering, may follow different trade-offs. We test one adversarial case, a detector that always names the wrong side, but not an attacker who controls which pairs are flagged.

Ethics Statement

This work uses public benchmarks and commercial model APIs, and involves no human subjects or personal data. Its main risk is dual use. Our results show that a false "likely misinformation" label, or the removal of a passage, can change a RAG system's answers: an inverted oracle that names the correct side as misinformation costs up to 21.8 points. An adversary who can trigger a deployed detector could exploit this (Zou et al., 2025). We therefore report harm on conflict-free questions alongside every gain, and we argue that systems should measure what a detector does at the document level before acting on its flags.

References

Akari Asai, Zeqiu Wu, Yizhong Wang, Avirup Sil, and Hannaneh Hajishirzi. 2024. Self-RAG: Learning to retrieve, generate, and critique through self-reflection. In *The Twelfth International Conference on Learning Representations (ICLR)*.

Dallas Card, Peter Henderson, Urvashi Khandelwal, Robin Jia, Kyle Mahowald, and Dan Jurafsky. 2020. With little power comes great responsibility. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 9263–9274, Online. Association for Computational Linguistics.

Arie Cattan, Alon Jacovi, Ori Ram, Jonathan Herzig, Roee Aharoni, Sasha Goldshtein, Eran Ofek, Idan Szpektor, and Avi Caciularu. 2025. DRAGged into conflicts: Detecting and addressing conflicting sources in search-augmented LLMs. *Preprint*, arXiv:2506.08500.

Hung-Ting Chen, Michael Zhang, and Eunsol Choi. 2022. Rich knowledge sources bring complex knowledge conflicts: Recalibrating models to reflect conflicting evidence. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pages 2292–2307, Abu Dhabi, United Arab Emirates. Association for Computational Linguistics.

Cheng-Han Chiang and Hung-yi Lee. 2024. Do metadata and appearance of the retrieved webpages affect LLM's reasoning in retrieval-augmented generation? In *Proceedings of the 7th BlackboxNLP Workshop: Analyzing and Interpreting Neural Networks for NLP*, pages 389–406, Miami, Florida, US. Association for Computational Linguistics.

Boyi Deng, Wenjie Wang, Fengbin Zhu, Qifan Wang, and Fuli Feng. 2025. CrAM: Credibility-aware attention modification in LLMs for combating misinformation in RAG. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 39, pages 23760–23768.

Rotem Dror, Gili Baumer, Segev Shlomov, and Roi Reichart. 2018. The hitchhiker's guide to testing statistical significance in natural language processing. In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1383–1392, Melbourne, Australia. Association for Computational Linguistics.

Bradley Efron and Robert J. Tibshirani. 1993. *An Introduction to the Bootstrap*, volume 57 of *Monographs on Statistics and Applied Probability*. Chapman & Hall, New York.

Ziyu Ge, Yuhao Wu, Daniel Wai Kit Chin, Roy Ka-Wei Lee, and Rui Cao. 2025. Resolving conflicting evidence in automated fact-checking: A study on retrieval-augmented LLMs. In *Proceedings of the Thirty-Fourth International Joint Conference on Artificial Intelligence, IJCAI-25*, pages 9656–9664. International Joint Conferences on Artificial Intelligence Organization.

Vignesh Gokul, Srikanth Tenneti, and Alwarappan Nakkiran. 2025. Contradiction detection in RAG systems: Evaluating LLMs as context validators for improved information consistency. *Preprint*, arXiv:2504.00180.

Stephanie Gross, Johann Petrak, and Brigitte Krenn. 2026. Ragability benchmark: A dataset and library to test LLMs on inter-context conflicts. In *Proceedings of the Fifteenth Language Resources and Evaluation Conference*, pages 2323–2333, Palma de Mallorca, Spain. ELRA Language Resource Association.

Suchin Gururangan, Swabha Swayamdipta, Omer Levy, Roy Schwartz, Samuel R. Bowman, and Noah A. Smith. 2018. Annotation artifacts in natural language inference data. In *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 2 (Short Papers)*, pages 107–112, New Orleans, Louisiana. Association for Computational Linguistics.

Pengcheng He, Jianfeng Gao, and Weizhu Chen. 2023. DeBERTaV3: Improving DeBERTa using ELECTRA-style pre-training with gradient-disentangled embedding sharing. In *The Eleventh International Conference on Learning Representations (ICLR)*.

Sture Holm. 1979. A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2):65–70.

Giwon Hong, Jeonghwan Kim, Junmo Kang, Sung-Hyon Myaeng, and Joyce Jiyoung Whang. 2024. Why so gullible? enhancing the robustness of retrieval-augmented models against counterfactual noise. In *Findings of the Association for Computational Linguistics: NAACL 2024*, pages 2474–2495, Mexico City, Mexico. Association for Computational Linguistics.

S M Asif Hossain, Ruksat Khan Shayoni, and M. F. Mridha. 2026. EvidentialRAG: Quantifying and mitigating information conflict in multi-source retrieval-augmented generation via evidential deep learning. *Preprint*, arXiv:2607.10491.

Yufang Hou, Alessandra Pascale, Javier Carnerero-Cano, Tigran Tchrakian, Radu Marinescu, Elizabeth Daly, Inkit Padhi, and Prasanna Sattigeri. 2024. WikiContradict: A benchmark for evaluating LLMs on real-world knowledge conflicts from Wikipedia. In *Advances in Neural Information Processing Systems (Datasets and Benchmarks Track)*, volume 37, pages 109701–109747.

Jeongyeon Hwang, Junyoung Park, Hyejin Park, Dongwoo Kim, Sangdon Park, and Jungseul Ok. 2025. Retrieval-augmented generation with estimation of source reliability. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 34279–34303, Suzhou, China. Association for Computational Linguistics.

Tomáš Kočiský, Jonathan Schwarz, Phil Blunsom, Chris Dyer, Karl Moritz Hermann, Gábor Melis, and Edward Grefenstette. 2018. The NarrativeQA reading comprehension challenge. *Transactions of the Association for Computational Linguistics*, 6:317–328.

Daniël Lakens. 2017. Equivalence tests: A practical primer for t tests, correlations, and meta-analyses. *Social Psychological and Personality Science*, 8(4):355–362.

Yoonsang Lee, Xi Ye, and Eunsol Choi. 2024. AmbigDocs: Reasoning across documents on different entities under the same name. In *First Conference on Language Modeling (COLM)*.

Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. 2020. Retrieval-augmented generation for knowledge-intensive NLP tasks. In *Advances in Neural Information Processing Systems*, volume 33, pages 9459–9474. Curran Associates, Inc.

Shayne Longpre, Kartik Perisetla, Anthony Chen, Nikhil Ramesh, Chris DuBois, and Sameer Singh. 2021. Entity-based knowledge conflicts in question answering. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pages 7052–7063, Online and Punta Cana, Dominican Republic. Association for Computational Linguistics.

Yifei Ming, Senthil Purushwalkam, Shrey Pandit, Zixuan Ke, Xuan-Phi Nguyen, Caiming Xiong, and Shafiq Joty. 2025. FaithEval: Can your language model stay faithful to context, even if "the moon is made of marshmallows". In *The Thirteenth International Conference on Learning Representations (ICLR)*.

OpenAI, Aaron Hurst, Adam Lerer, Adam P. Goucher, Adam Perelman, Aditya Ramesh, et al. 2024. GPT-4o system card. *Preprint*, arXiv:2410.21276.

Liangming Pan, Wenhu Chen, Min-Yen Kan, and William Yang Wang. 2023a. Attacking open-domain question answering by injecting misinformation. In *Proceedings of the 13th International Joint Conference on Natural Language Processing and the 3rd Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 525–539, Nusa Dua, Bali. Association for Computational Linguistics.

Ruotong Pan, Boxi Cao, Hongyu Lin, Xianpei Han, Jia Zheng, Sirui Wang, Xunliang Cai, and Le Sun. 2024. Not all contexts are equal: Teaching LLMs credibility-aware generation. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 19844–19863, Miami, Florida, USA. Association for Computational Linguistics.

Yikang Pan, Liangming Pan, Wenhu Chen, Preslav Nakov, Min-Yen Kan, and William Wang. 2023b. On the risk of misinformation pollution with large language models. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 1389–1403, Singapore. Association for Computational Linguistics.

Qwen Team, An Yang, Baosong Yang, et al. 2024. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*.

Nils Reimers and Iryna Gurevych. 2019. Sentence-BERT: Sentence embeddings using Siamese BERT-networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 3982–3992, Hong Kong, China. Association for Computational Linguistics.

Jakob Schuster, Vagrant Gautam, and Katja Markert. 2026. Whose facts win? LLM source preferences under knowledge conflicts. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 29430–29459, San Diego, California, United States. Association for Computational Linguistics.

Freda Shi, Xinyun Chen, Kanishka Misra, Nathan Scales, David Dohan, Ed H. Chi, Nathanael Schärli, and Denny Zhou. 2023. Large language models can be easily distracted by irrelevant context. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pages 31210–31227. PMLR.

Zhaochen Su, Jun Zhang, Xiaoye Qu, Tong Zhu, Yanshu Li, Jiashuo Sun, Juntao Li, Min Zhang, and Yu Cheng. 2024. ConflictBank: A benchmark for evaluating the influence of knowledge conflicts in LLMs. In *Advances in Neural Information Processing Systems (Datasets and Benchmarks Track)*, volume 37, pages 103242–103268.

Emiel van Miltenburg, Chris van der Lee, and Emiel Krahmer. 2021. Preregistering NLP research. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 613–623, Online. Association for Computational Linguistics.

Chenyu Wang, Yueyuan Li, Yingmin Liu, and Yang Shu. 2026. ConflictRAG: Detecting and resolving knowledge conflicts in retrieval augmented generation. *Preprint*, arXiv:2605.17301.

Fei Wang, Xingchen Wan, Ruoxi Sun, Jiefeng Chen, and Sercan O Arik. 2025a. Astute RAG: Overcoming imperfect retrieval augmentation and knowledge conflicts for large language models. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 30553–30571, Vienna, Austria. Association for Computational Linguistics.

Han Wang, Archiki Prasad, Elias Stengel-Eskin, and Mohit Bansal. 2025b. Retrieval-augmented generation with conflicting evidence. In *Second Conference on Language Modeling (COLM)*.

Yike Wang, Shangbin Feng, Heng Wang, Weijia Shi, Vidhisha Balachandran, Tianxing He, and Yulia Tsvetkov. 2024. Resolving knowledge conflicts in large language models. In *First Conference on Language Modeling*.

Zhepei Wei, Wei-Lin Chen, and Yu Meng. 2025. InstructRAG: Instructing retrieval-augmented generation via self-synthesized rationales. In *The Thirteenth International Conference on Learning Representations*.

Orion Weller, Aleem Khan, Nathaniel Weir, Dawn Lawrie, and Benjamin Van Durme. 2024. Defending against disinformation attacks in open-domain question answering. In *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics (Volume 2: Short Papers)*, pages 402–417, St. Julian's, Malta. Association for Computational Linguistics.

Kevin Wu, Eric Wu, and James Zou. 2024. ClashEval: Quantifying the tug-of-war between an LLM's internal prior and external evidence. In *Advances in Neural Information Processing Systems (Datasets and Benchmarks Track)*, volume 37, pages 33402–33422.

Chong Xiang, Tong Wu, Zexuan Zhong, David Wagner, Danqi Chen, and Prateek Mittal. 2024. Certifiably robust RAG against retrieval corruption. *Preprint*, arXiv:2405.15556.

Jian Xie, Kai Zhang, Jiangjie Chen, Renze Lou, and Yu Su. 2024. Adaptive chameleon or stubborn sloth: Revealing the behavior of large language models in knowledge conflicts. In *The Twelfth International Conference on Learning Representations (ICLR)*.

Rongwu Xu, Zehan Qi, Zhijiang Guo, Cunxiang Wang, Hongru Wang, Yue Zhang, and Wei Xu. 2024. Knowledge conflicts for LLMs: A survey. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 8541–8565, Miami, Florida, USA. Association for Computational Linguistics.

Shi-Qi Yan, Jia-Chen Gu, Yun Zhu, and Zhen-Hua Ling. 2024. Corrective retrieval augmented generation. *Preprint*, arXiv:2401.15884.

Hua Ye, Siyuan Chen, Ziqi Zhong, Canran Xiao, Haoliang Zhang, Yuhan Wu, and Fei Shen. 2026. Seeing through the conflict: Transparent knowledge conflict handling in retrieval-augmented generation. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 40, pages 34423–34431.

Yizhen Yuan, Rui Kong, Dongze Li, Yuanchun Li, and Yunxin Liu. 2026. Benchmarking LLM's capability in reasoning over conflicting web references. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 303–322, San Diego, California, United States. Association for Computational Linguistics.

Huichi Zhou, Kin-Hei Lee, Zhonghao Zhan, Yue Chen, Zhenhao Li, Zhaoyang Wang, Hamed Haddadi, and Emine Yilmaz. 2025. TrustRAG: Enhancing robustness and trustworthiness in retrieval-augmented generation. *arXiv preprint arXiv:2501.00879*.

Wei Zou, Runpeng Geng, Binghui Wang, and Jinyuan Jia. 2025. PoisonedRAG: Knowledge corruption attacks to retrieval-augmented generation of large language models. In *34th USENIX Security Symposium (USENIX Security 25)*, pages 3827–3844, Seattle, WA. USENIX Association.

A Prompts

All prompts below are copied verbatim from the code that produced the results.

Generation skeleton (default).

```
Answer the question using the passages below.
If the passages support more than one valid answer
(for example, different people or things share the
same name), give every valid answer.
If no passage answers the question, the answer is
"unknown".
{instruction}
{context}
{conflicts}
Question: {question}

Give a brief explanation, then end with one line in
exactly this format:
Final answer(s): <answer 1> | <answer 2> | ...
```

Minimal skeleton (no answer policy).

```
Answer the question using the passages below.
If no passage answers the question, the answer is
"unknown".
{instruction}
{context}
{conflicts}
Question: {question}

Give a brief explanation, then end with one line in
exactly this format:
Final answer(s): <your answer; if there is more than
one, separate them with |>
```

Instruction v1.

```
Some passages may contradict each other. Before
answering, check whether they do.
Ignore passages that appear to contain
misinformation, and if a disagreement reflects
genuinely different valid answers, report each one.
```

Instruction v2.

```
Some passages may contain false information. Before
answering, decide which passages are reliable by
checking them against each other.
Do not include an answer that is supported only by a
passage you judge to be unreliable.
```

Example: directional flag (passage 2 suspect).

```
Answer the question using the passages below.
If the passages support more than one valid answer
(for example, different people or things share the
same name), give every valid answer.
If no passage answers the question, the answer is
"unknown".

Passage 0 [POTENTIAL CONFLICT with passage(s): 2]:
<passage 0>

Passage 1:
```

```
<passage 1>

Passage 2 [LIKELY MISINFORMATION; conflicts with
passage(s): 0]:
<passage 2>

POTENTIAL CONFLICTS DETECTED:
- Passages 0 and 2: possible contradiction; passage
2 likely contains misinformation

Question: Who founded X?

Give a brief explanation, then end with one line in
exactly this format:
Final answer(s): <answer 1> | <answer 2> | ...
```

Directional LLM detector.

```
You will see a question and {n} numbered passages
retrieved for it.
Some passages may contain misinformation: false
claims about the answer to the question.
Identify every pair of passages that CONTRADICT each
other about the answer to the question,
i.e. they cannot both be true about the same entity
at the same time. Passages that are about
different entities with the same name, give
complementary details, or describe different times
are NOT contradictions; note them as "ambiguity",
"complementary" or "temporal" instead.

For each contradiction, say which passage of the
pair is more likely the misinformation, judging
by agreement with the other passages and by
plausibility: its number, "both" if both are likely
false, or "unclear" if you cannot tell.

Question: {question}

{passages}

Return JSON only: {{"pairs": [{{"i": <int>, "j":
<int>, "relation":
"contradiction|ambiguity|complementary|temporal",
"misinformation": <int>|"both"|"unclear"}}]}}
Use the passage numbers shown. "misinformation" is
needed only for contradictions. Return {{"pairs":
[]}} if there are none.
```

B Additional Results

Robustness to the metric and the labels. Table 7 repeats Table 1 with the bidirectional metric. Table 8 repeats the pre-registered tests with every suspect document treated as misinformation. Table 5 lists all pre-registered tests.

The heuristic detector. CAAG is the detector we set out to study. As a neutral flag it did not beat random flags of the same volume, and as a removal rule it only deleted correct passages (-7.9 points on clean questions, 0.0 on conflict ones). Its false flags fall mostly on pairs that should both be reported: 134 of its 221 false flags on the hold-out are same-name ambiguity pairs, which its entity-mismatch signal flags because they mention different dates or numbers. We read this as a warning about lexical contradiction cues in general (Gururangan et al., 2018), rather than as a problem with a particular threshold. Retuning illustrates why F1 misleads. Over a 121-setting sweep, CAAG's best pair F1 is 29.2, above NLI's, at $\theta_s = 0.40$ and $\theta_c = 0.10$. But it gets there by flagging 43% of all pairs and 87% of clean questions, which our grid predicts would be harmful under removal (Appendix B).

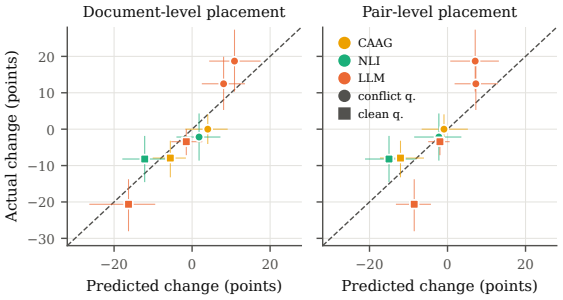


Figure 1: Effect of real detectors predicted from the synthetic grid (x, with the 95% bootstrap prediction interval) against the measured effect (y, with its 95% CI), for conflict (●) and clean (■) questions. Left: detectors located by their document-level behaviour; right: by pair precision, recall and direction accuracy.

Original CAAG format. Rendering CAAG's flags in its original format, with typed lines and confidence scores ("factual conflict (confidence: 0.41)"), changes accuracy by -0.9 points overall (-1.2 on conflict questions, -1.6 on clean ones). This is no different from the uniform format.

Pilot study (exploratory). On the 200-question pilot split, scored with the bidirectional metric, the same pattern appeared before any confirmatory run. Oracle neutral flags changed conflict-question accuracy by -10.5 points, directional flags by $+11.8$, and removing the suspect by $+28.4$. The pilots also showed that the original REMOVE strategy, which deletes an arbitrary side of each pair, removes correct documents. They are why we added directional formats, synthetic direction accuracy and document-level placement. All pilot runs are released.

Phase-0 audit of CAAG on NarrativeQA. CAAG was designed for long-document QA. On 100 NarrativeQA (Kočiský et al., 2018) questions from 50 stories (BM25 top-10 chunks from a single coherent story), it flags at least one pair in 54% of questions, 166 pairs in all. A single story rarely

contradicts itself, so most of these flags are likely false; the flagged pairs are released for annotation. This is the setting where a false flag can only hurt.

C Further Results

C.1 Where the directional flag fails (exploratory)

After seeing the minimal-skeleton run, we split conflict questions by the number of gold answers. 105 of 139 have several (same-name ambiguity *and* misinformation), and a strict answer must list all of them. On the 34 single-answer conflict questions, the directional flag's estimate is positive in every setting, from $+11.8$ points (Qwen) to $+46.1$ (gpt-4o-mini), both with the default skeleton. On multi-answer questions it changes little except with gpt-4.1 (gpt-4o-mini $+2.9$, and -5.7 with the minimal skeleton; gpt-4.1 $+19.0$). One reading is that a warning about one passage makes the smaller models more conservative about the legitimate alternative answers too; we did not test this. Removal does not depend on the model acting on a warning, and it gains in both groups.

C.2 Real detectors

CAAG flags 34.1% of clean questions. Its neutral flags do not beat random flags at the same rate (H5; -0.4 vs. $+1.0$ points overall). The 90% CI of the difference, $[-3.1, +0.1]$, rules out a gain larger than $+0.1$ points but does not establish equivalence within ± 3 points (TOST). The directional-versus-neutral comparison with the LLM detector's pairs was registered as exploratory in the addendum. Under the bidirectional metric, the LLM detector's directional flag costs clean questions 6.1 points. With gpt-4.1 as the generator, its gain over all 300 questions ($+5.4$) is significant after Holm, but the 35 unlabelled questions contribute disproportionately to it (they gain 13.3 points); over the labelled questions it is $+4.4$ $[+0.6, +8.3]$. On ConflictBank the NLI cross-encoder flags no pair at all: ConflictBank passages are long, LLM-written and agree on everything but the one fact, and the truncated pair rarely contains it.

C.3 A detector from another family

Replacing the gpt-4o-mini detector with Qwen2.5-7B (gpt-4o-mini still generates) gives a much less selective detector. It flags 96.8% of clean RAMDocs questions, but its direction accuracy is 89.0%. Its directional flag gains $+10.6$ points on conflict questions and 0.0 on clean ones, and its neutral flag -2.6. Removal again trades a conflict-question gain ($+14.1$) for a clean-question loss (-14.6). The conclusions do not depend on the detector's family.

C.4 CONFLICTS

On the 237 automatically scorable CONFLICTS questions (exploratory, one seed; most have no conflict), no condition moves accuracy significantly (Table 10).

D Reproducibility

Runs. Every number in this paper is produced by one script from committed run outputs: per-call responses, token counts, detector outputs and configurations. The pre-registration was committed before any confirmatory run. Its deviations are listed below.

Deviations from the pre-registration.

1. *Primary metric.* The first draft of the pre-registration named the bidirectional metric as primary. Before the confirmatory run we switched to one-way matching, because it reproduces MADAM-RAG's published score on the pilot split (Section 4); bidirectional became the robustness metric. Every table reports both.

2. *Second generator.* The plan named an open-weights model. None was available before the confirmatory run, so gpt-4.1 was used (H9). After review we added Qwen2.5-7B-Instruct, registered before its run. Because local inference takes about 3 s per call, the Qwen runs use one seed; we made that change before inspecting any Qwen result.

3. *CONFLICTS judge.* The plan called for a judge from a different model family, validated on about 100 human labels. Neither the judge nor the labels were available, so CONFLICTS is scored only on items with a reference answer, with one seed, and reported as exploratory.

4. *H4.* The pre-registration counted 54 false-flag cells. On clean questions, cells that differ only in direction accuracy render identical prompts (their scores differ by exactly 0), so they are one comparison: 18 in all, 9 per format. We report H4a per format. The cell-permutation p-value for the dose-response correlation assumed exchangeable cells; the cells are nested, so we re-

place it with a question-level bootstrap CI. No threshold was set for H4c, so we report a count.

5. *H5.* A non-significant difference is not equivalence. We add a TOST with a ± 3-point margin, chosen in the addendum after the confirmatory run, so it is post hoc.

6. *H6.* The first draft tested whether the grid prediction lies inside the actual effect's CI, which is lenient. We now report the pre-registered test (actual effect inside a bootstrap prediction interval) and a paired test of the two spaces' errors.

7. *Placement modes.* For H6, the detectors without direction were run under the label `remove_suspect` as well as `remove`. Both render identical prompts, so these were served entirely from the response cache.

8. *Significance.* Earlier drafts used different corrections in different tables. We now apply one scheme throughout (Section 4), which makes some exploratory differences that were bold in earlier drafts non-significant.

9. *Single- and multi-answer split.* The split of conflict questions by number of gold answers (Appendix C.1) was chosen after seeing the minimal-skeleton run and is exploratory.

10. *Second registration.* After a second review we registered H12–H14 (ConflictBank, the neutral flag with a resolution instruction, and the published baselines), the Qwen2.5-7B detector and the ConflictBank sampling rule. The gpt-4.1 runs in it use seed 0 only, to stay within budget. One ConflictBank gpt-4o-mini run stopped on a transient API error and was resumed from the response cache; no row was re-sampled.

The addendum (H10, H11 and the re-analyses above), the Qwen seed change and the second registration were committed to the repository before the runs they cover. The repository's commit history is the only timestamp; we did not use a third-party registry (van Miltenburg et al., 2021).

Metric choice. With the one-way metric, our port of MADAM-RAG scores 28.5% on the pilot split and 32.8% on all 500 questions (published: 28.00, on an undisclosed subset); with the bidirectional metric it scores 38.4% on all 500. Because the published subset and scorer are unknown, we do not claim to reproduce the published number. Normalisation lower-cases and removes punctuation and articles.

WikiContradict and CONFLICTS. A WikiContradict answer is correct if both of the dataset's answer strings (one per passage) match a final answer under the bidirectional rule. From CONFLICTS we use the 237 questions that have a reference answer (outdated-information, misinformation and no-conflict items); an answer is correct if it contains the reference answer.

Baselines. MADAM-RAG is ported unchanged to our client: three debate rounds, temperature 0, 128 tokens per call. Astute RAG has no official code. We use the prompts in its appendix verbatim with its default settings (t=1 consolidation round, m=1 generated passage, temperature 0, 1,024 output tokens). A generated passage that says "I don't know" is dropped, and the answer is the text in its `<ANSWER>` tag. TrustRAG is ported from its repository (commit `11dcea0`, MIT licence): SimCSE CLS embeddings (`sup-simcse-bert-base-uncased`), a ROUGE-L gate of 0.25, 2-means clustering and a cosine threshold of 0.88, followed by its three prompts verbatim at temperature 0.1. Its system message is not sent, because our client sends user messages only. The *multi* variants change one sentence of the final step to ask for every valid answer (Astute RAG) or append our final-answer line (TrustRAG); the unmodified variants are also reported.

ConflictBank. We use the QA file of the dataset's Hugging Face release (revision `92866a0`). Questions are sampled round-robin over relations with seed 0; distractors are other subjects' correct passages. The sampling script, the item ids and the seed are released; the data are not redistributed, because the release states a licence in its README but ships no licence file.

Cost and compute. All API calls in this paper, pilots included, cost $51.75. Prices were checked on 2026-10-27 (gpt-4o-mini: $0.15, $0.075 cached, and $0.60 per million input, cached-input and output tokens; gpt-4.1: $2.00, $0.50 and $8.00). CAAG and NLI ran on a laptop CPU (0.2 s and 1.2 s per question).

E Synthetic Detectors and the Grid

Figure 3 replaces the oracle with synthetic detectors of known quality. Two thresholds appear. First, *direction accuracy*: at a direction accuracy of 0.5 or 0.75, no cell gains significantly on all

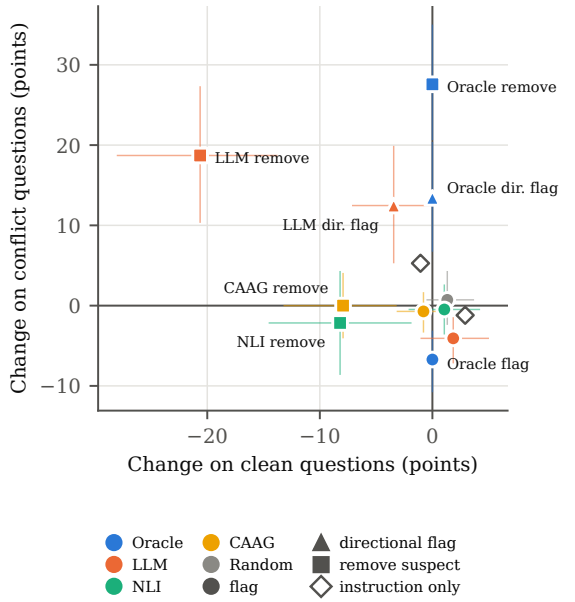


Figure 2: Change in one-way strict accuracy on RAMDocs conflict questions (y) against clean questions (x), relative to no flags (gpt-4o-mini, default skeleton, confirmatory split, 95% CIs). Colour: detector; shape: format. The oracle sits on the y-axis because it never flags a clean question. Without direction, removal deletes the second passage of each pair.

questions in either format. Only perfect direction does (directional flag +6, removal +13 points at perfect precision). Second, *precision*, but mainly for removal. On clean questions a cell's prompt does not depend on direction accuracy, so the false-flag cells reduce to 9 distinct comparisons per format. Removal is significantly harmful in 6 of them, the directional flag in 0 (H4a): false flags make the model delete correct passages, which costs clean questions 15 points at precision 0.5 and 26 at 0.25, while synthetic false directional flags cost at most two points. The harm tracks the share of clean questions that lose a correct document (Spearman ρ =-0.39 over 24 distinct cells, 95% CI [−0.74, −0.12] from a question-level bootstrap; H4b). Removal hurt clean questions significantly more than the directional flag in 5 of 9 matched cells (H4c; we set no threshold, so we report the count). Removal is the more powerful and the more fragile interface: its best cell gains twice the directional flag's on all questions, but below precision 0.75 it is the only format that does net harm.

Does the grid predict real detectors? If the grid only described synthetic detectors, it would be of limited use. Figure 1 (Appendix B) tests whether it predicts real ones (H6). As preregistered, we bootstrap the whole prediction (cell effects, every detector's location and the interpolation) over questions, and ask whether the actual effect falls inside the 95% interval. Locating detectors by what they do to documents (the share of conflict questions whose misinformation is marked, and the share of questions that have a correct document marked) puts 7 of 8 effects inside; locating them by pair precision, recall and direction accuracy puts 6 inside. Document-level errors are smaller on average (4.1 vs. 5.3 points), but the paired difference, −1.2 [−3.6, +0.4] (exact sign-flip $p = 0.41$), is not significant over 8 cases. Both spaces miss the same case, the LLM detector's removal gain on conflict questions: one correctly directed pair is enough to remove a misinformation document, so the detector removes the misinformation on 61.2% of conflict questions despite its low pair recall. We chose the document-level axes after seeing the LLM detector on the pilot split. The grid is a useful first estimate of a detector's effect, but with three detectors we cannot claim that one summary predicts better than the other.

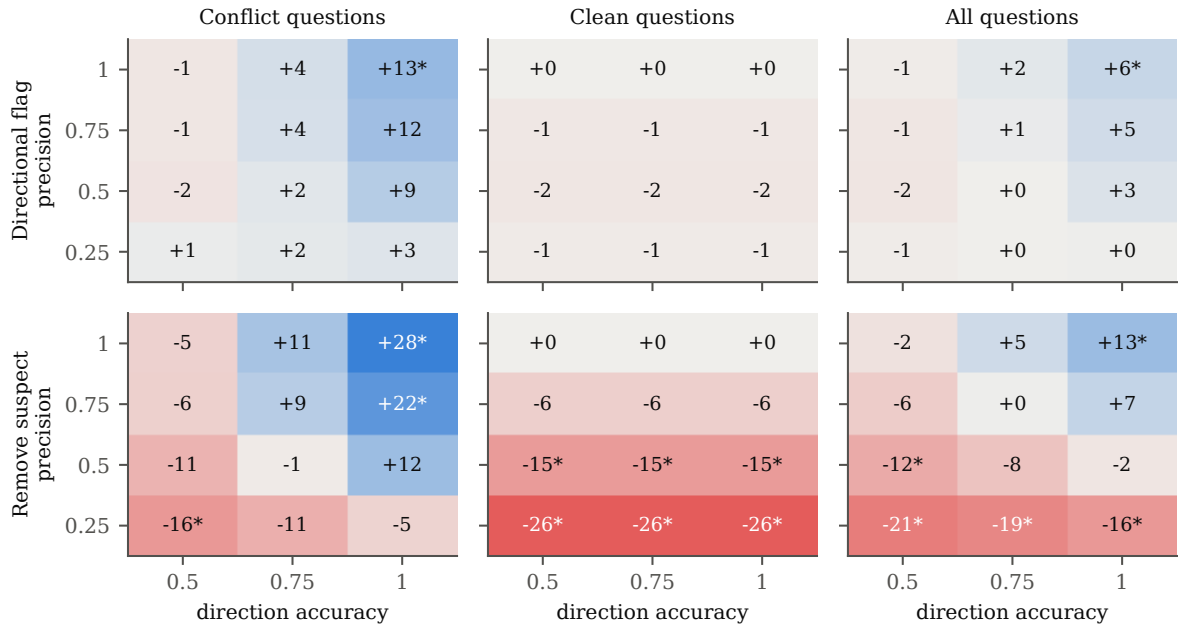


Figure 3: Synthetic detectors at recall 1: change in one-way strict accuracy (points) against no flags, by pair precision (rows) and direction accuracy (columns), for the directional flag (top) and removal (bottom). *: significant after Holm correction over the 36 cells of that mode and subset. Clean questions only see false flags, so their rows do not vary with direction accuracy.

	Pre-registered test	Estimate [95% CI]	p_{Holm}	Outcome	Robust
H1	F1 llm_dir > nli	+2.6 [−2.4, +7.5]	0.311	not supported	–
H1	F1 nli > caag	+12.2 [+7.5, +16.9]	<0.001	supported	–
H1	direction accuracy llm_dir > 50% (cluster CI[r])	88.3% [83.3, 92.9]	<0.001	supported	–
H2	oracle+flag < plain (conflict)	−6.7 [−10.8, −3.1]	<0.001	supported	yes
H3	oracle+flag_directional > plain (conflict)	+13.4 [+6.7, +20.4]	<0.001	supported	yes
H3	oracle+flag_directional > oracle+flag (conflict)	+20.1 [+13.9, +26.9]	<0.001	supported	yes
H3	oracle+remove_suspect > plain (conflict)	+27.6 [+20.6, +35.0]	<0.001	supported	yes
H4a	false-flag cells below plain on clean q., directional flag[r]	0 of 9 cells	Holm	not supported	yes
H4a	false-flag cells below plain on clean q., removal[r]	6 of 9 cells	Holm	supported	yes
H4b	harm grows with clean-q. exposure, Spearman over 24 cells[r]	$\rho = -0.39$ [−0.74, −0.12]	–	supported	yes
H4c	removal harms clean q. more than directional flag[r]	5 of 9 matched cells	Holm	count only	–
H5	caag+flag vs random+flag (all)	−1.4 [−3.4, +0.3]	0.178	n.s.	yes
H5	equivalence within ±3 points (TOST)[r]	90% CI [−3.1, +0.1]	–	not equivalent	yes
H6	actual effect inside the 95% interval of the document-level prediction[r]	7 of 8 (pair level: 6 of 8)	–	descriptive	–
H6	document-level error < pair-level error (paired, exact sign-flip)[r]	−1.2 [−3.6, +0.4]	0.414	not supported	yes
H7	instruct vs plain (conflict)	−1.2 [−4.3, +1.7]	1.000	n.s.	yes
H7	instruct vs plain (clean)	+2.9 [0.0, +6.1]	0.280	n.s.	yes
H7	instruct_v2 vs plain (conflict)	+5.3 [+1.4, +9.6]	0.059	n.s.	yes
H7	instruct_v2 vs plain (clean)	−1.1 [−5.0, +2.6]	1.000	n.s.	yes
H9	gpt-4.1: oracle+flag < plain (conflict)	−8.6 [−13.2, −4.3]	<0.001	supported	yes
H9	gpt-4.1: oracle+flag_directional > plain (conflict)	+18.2 [+12.0, +24.7]	<0.001	supported	yes
H9	gpt-4.1: oracle+flag_directional > oracle+flag (conflict)	+26.9 [+20.1, +33.6]	<0.001	supported	yes
H9	gpt-4.1: oracle+remove_suspect > plain (conflict)	+30.2 [+22.5, +37.9]	<0.001	supported	yes
H8	WikiContradict: instruct vs plain	+8.7 [+5.4, +12.3]	<0.001	significant	–
H8	WikiContradict: instruct_v2 vs plain	−6.7 [−10.0, −3.7]	<0.001	significant	–
H8	WikiContradict: oracle+flag vs plain	+6.5 [+3.0, +10.0]	<0.001	supported	–
H8	WikiContradict: caag+flag vs plain	+0.7 [0.0, +1.6]	0.250	n.s.	–
H8	WikiContradict: nli+flag vs plain	+3.2 [+0.3, +6.1]	0.077	n.s.	–
H8	WikiContradict: llm_dir+flag vs plain	+6.2 [+2.8, +9.6]	<0.001	significant	–
H10	gpt-4o-mini: oracle+flag vs plain (conflict)	−5.3 [−9.4, −1.4]	0.042	significant	yes
H10	gpt-4o-mini: oracle+flag_directional > plain (conflict)	+5.0 [−1.7, +11.8]	0.314	not supported	yes
H10	gpt-4o-mini: oracle+flag_directional > oracle+flag (conflict)	+10.3 [+4.8, +16.1]	0.005	supported	yes
H10	gpt-4o-mini: oracle+remove_suspect > plain (conflict)	+18.9 [+12.5, +25.9]	0.001	supported	yes
H10	gpt-4.1: oracle+flag vs plain (conflict)	−11.0 [−17.5, −4.6]	0.010	significant	yes
H10	gpt-4.1: oracle+flag_directional > plain (conflict)	+12.7 [+5.8, +19.7]	0.005	supported	yes
H10	gpt-4.1: oracle+flag_directional > oracle+flag (conflict)	+23.7 [+17.0, +30.7]	0.001	supported	yes
H10	gpt-4.1: oracle+remove_suspect > plain (conflict)	+14.9 [+8.6, +21.6]	0.001	supported	yes
H10	Qwen2.5-7B: oracle+flag vs plain (conflict)	−5.8 [−10.8, −1.4]	0.117	n.s.	yes
H10	Qwen2.5-7B: oracle+flag_directional > plain (conflict)	+2.9 [−3.6, +9.4]	0.509	not supported	yes
H10	Qwen2.5-7B: oracle+flag_directional > oracle+flag (conflict)	+8.6 [+2.9, +14.4]	0.041	supported	yes
H10	Qwen2.5-7B: oracle+remove_suspect > plain (conflict)	+10.8 [+4.3, +17.3]	0.017	supported	yes
H11	Qwen2.5-7B: oracle+flag < plain (conflict)	−5.8 [−10.8, −0.7]	0.114	not supported	yes
H11	Qwen2.5-7B: oracle+flag_directional > plain (conflict)	+3.6 [−0.7, +8.6]	0.224	not supported	yes
H11	Qwen2.5-7B: oracle+flag_directional > oracle+flag (conflict)	+9.4 [+5.0, +14.4]	0.001	supported	yes
H11	Qwen2.5-7B: oracle+remove_suspect > plain (conflict)	+18.0 [+10.8, +25.2]	<0.001	supported	yes
H12	gpt-4o-mini: neutral flag vs plain (conflict)	−8.8 [−12.8, −4.9]	0.001	significant	yes
H12	gpt-4o-mini: dir. flag > plain (conflict)	+27.9 [+22.8, +32.9]	0.001	supported	yes
H12	gpt-4o-mini: dir. flag > neutral flag (conflict)	+36.7 [+31.6, +41.9]	0.001	supported	yes
H12	gpt-4o-mini: removal > plain (conflict)	+66.4 [+61.1, +71.7]	0.001	supported	yes
H12	gpt-4.1: neutral flag vs plain (conflict)	−6.3 [−9.7, −3.3]	0.001	significant	yes
H12	gpt-4.1: dir. flag > plain (conflict)	+23.0 [+18.0, +28.3]	0.001	supported	yes
H12	gpt-4.1: dir. flag > neutral flag (conflict)	+29.3 [+24.0, +34.7]	0.001	supported	yes
H12	gpt-4.1: removal > plain (conflict)	+78.3 [+73.3, +83.0]	0.001	supported	yes
H12	Qwen2.5-7B: neutral flag vs plain (conflict)	−8.3 [−12.3, −4.3]	0.001	significant	yes
H12	Qwen2.5-7B: dir. flag > plain (conflict)	+17.7 [+12.0, +23.3]	0.001	supported	yes
H12	Qwen2.5-7B: dir. flag > neutral flag (conflict)	+26.0 [+20.7, +31.3]	0.001	supported	yes
H12	Qwen2.5-7B: removal > plain (conflict)	+63.7 [+58.3, +69.0]	0.001	supported	yes
H13	RAMDocs, gpt-4o-mini: neutral flag + instruction vs plain (conflict)	−2.4 [−6.2, +1.2]	0.743	n.s.	yes
H13	RAMDocs, gpt-4o-mini: dir. flag + instruction > neutral flag + instruction (conflict)	+15.6 [+9.6, +21.8]	<0.001	supported	yes

	Pair level			Clean-q.	Direc-	Removal, per question		Latency	Cost
Detector	P	R	F1	flagged	tion	misinfo gone	correct lost	(s)	($/1k q.)
CAAG (heuristic)	21.7	7.9	11.6	34.1	–	9.4	31.7	0.23	0.000
NLI cross-encoder	20.1	29.3	23.8	61.9	–	44.6	52.4	1.18	0.000
LLM (gpt-4o-mini)	37.2	20.5	26.4	76.2	88.3	61.2	65.9	0.80	0.181

Table 6: Detectors on the RAMDocs hold-out split (300 questions; percentages). *Clean-q. flagged*: share of conflict-free questions with at least one flag. *Direction*: share of correctly flagged contradictions whose suspect side is the misinformation. *Misinfo gone*: conflict questions whose misinformation documents are all removed when removing the suspect (or, without direction, the second passage). *Correct lost*: clean questions that lose at least one correct document. Latency is per question; CAAG and NLI run locally.

Condition	All	Lab.	Conflict	Clean	Tokens
No flags (plain)	35.2	37.2	25.9	49.7	911
Prompt only					
Instruction (v1)	+1.3	+0.5	−1.0	+2.1	948
Instruction (v2)	+2.2	+1.4	+3.6	−1.1	951
Oracle detector					
Oracle, flag	**−3.6**	**−4.0**	**−7.7**	0.0	980
Oracle, directional flag	+7.0	**+7.9**	**+15.1**	0.0	998
Oracle, remove suspect	**+15.4**	**+17.5**	**+33.3**	0.0	825
Real detectors (exploratory)					
Random, flag	+0.4	+0.4	−0.2	+1.1	949
CAAG, flag	−0.9	−1.4	−1.9	−0.8	942
CAAG, remove	−4.0	−4.5	−1.0	−8.5	809
NLI, flag	+0.2	0.0	−1.4	+1.6	1021
NLI, remove	−8.0	−7.9	−3.1	−13.2	592
LLM, flag	−3.4	−3.4	**−7.2**	+0.8	970
LLM, directional flag	+4.7	+4.5	+14.1	−6.1	983
LLM, remove suspect	−2.1	−2.6	**+18.9**	**−26.5**	718
Reference					
MADAM-RAG	+5.7	+5.2	+3.6	+6.9	8196

Table 7: RAMDocs hold-out, gpt-4o-mini, 3 seeds, bidirectional strict accuracy (robustness). *All*: 300 questions, including 35 whose conflict status the label rule leaves open; *Lab.*: the 265 labelled ones (139 conflict, 126 clean). First row: accuracy (%) with no flags. Other rows: paired difference from it in points; **bold** = significant after Holm correction across the table (sign-flip test, $p < 0.05$). Real-detector rows were not pre-registered as tests. Tokens: mean prompt tokens per call (MADAM-RAG: per question, all calls).

	Test (labels: *contains*)	One-way [95% CI]	p_{Holm}	Bidir.	p_{Holm}
H2	oracle+flag < plain (conflict)	−4.6 [−8.1, −1.5]	0.006	−5.2	0.003
H3	oracle+flag_directional > plain (conflict)	+11.6 [+5.6, +17.3]	<0.001	+13.3	<0.001
H3	oracle+flag_directional > oracle+flag (conflict)	+16.2 [+10.8, +21.8]	<0.001	+18.5	<0.001
H3	oracle+remove_suspect > plain (conflict)	+25.8 [+19.5, +32.2]	<0.001	+30.3	<0.001
H4	harmful cells, clean q. (dir. flag; removal); ρ	0/9; 8/9; $\rho = -0.08$ [-0.56, -0.02]			

Table 8: Label robustness: pre-registered tests with every suspect document treated as misinformation (173 conflict and 127 clean hold-out questions).

CAAG setting	θ_s	θ_c	P	R	F1	Clean FPR
Default	0.65	0.35	21.7	7.9	11.6	34.1
Best pair F1	0.40	0.10	20.7	49.6	29.2	87.3
Best question F1	0.55	0.15	22.2	28.8	25.1	60.3

Signals (defaults)	Pair F1	Flag rate	Clean FPR
all signals	11.6	6.2	34.1
negation only	0.6	0.4	4.0
entity only	8.6	4.3	23.0
temporal only	0.0	0.0	0.0

Table 9: CAAG on the RAMDocs hold-out (%). Top: thresholds from a sweep over 121 settings ($\theta_s \in [0.40, 0.90]$, $\theta_c \in [0.10, 0.60]$). Bottom: each signal alone.

Condition	All	Conflict	No conflict
No flags (plain)	66.7	49.3	73.8
Instruction (v2)	−0.4	+1.4	−1.2
CAAG, flag	+0.4	+1.4	0.0
NLI, flag	−1.7	−4.3	−0.6
LLM, flag	+1.7	+2.9	+1.2
LLM, directional flag	−0.4	−2.9	+0.6

Table 10: CONFLICTS, 237 automatically scorable questions (69 conflict), gpt-4o-mini, one seed; exploratory. Contains-match accuracy (%) and differences in points; no difference is significant (bold would mark raw $p < 0.05$). CAAG: flags 94.2% of conflict and 92.9% of clean questions; NLI: flags 52.2% of conflict and 43.5% of clean questions; LLM: flags 98.6% of conflict and 83.9% of clean questions.

Condition	All	Lab.	Conflict	Clean
No flags (plain)	32.7	34.5	21.8	48.4
Instruction (v2)	+5.8	+5.2	+12.2	−2.6
Oracle, flag	**−4.0**	**−4.5**	**−8.6**	0.0
Oracle, directional flag	**+8.4**	**+9.6**	**+18.2**	0.0
Oracle, remove suspect	**+14.0**	**+15.8**	**+30.2**	0.0
LLM, directional flag	+5.4	+4.4	+8.4	0.0
LLM, remove suspect	+1.0	−0.8	**+20.6**	**−24.3**
NLI, remove	−7.0	−7.3	+0.2	**−15.6**

Table 11: Replication with gpt-4.1 as generator (RAMDocs hold-out, 3 seeds, one-way strict accuracy; detector fixed to gpt-4o-mini). Same conventions as Table 1.

Condition	Both answers
No flags (plain)	14.0
Instruction (v1)	**+8.7**
Instruction (v2)	**−6.7**
Oracle, flag	**+6.5**
CAAG, flag	+0.7
NLI, flag	+3.2
LLM, flag	**+6.2**

Table 12: WikiContradict (253 genuine conflicts between two Wikipedia passages; the reference answer is both sides), gpt-4o-mini, 3 seeds. First row: share of answers giving both sides (%); other rows: difference in points, **bold** = significant after Holm within H8.